MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPES 5839 AND 5852

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tubes described heren shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Full wave, high-vacuum rectifier

Outline	---	See figure 1
Base	---	Intermediate shell octal, 6-pin (Melamine with barriers-space between base and bulb filled with potting compound)
Envelope	---	T9
Cathode	---	Coated unipotential
Heaters	---	Tabbed with nickel sleeves at point of weld to stem leads
Heater insulator	---	Extruded aluminum oxide - mandatory
Stem	---	8-wire button - lead glass

Base connections:

Pin No.	---	1	2	3	5	7	8
Element	---	---	h	2a	1a	h	k

(Do not connect leads to pin No. 1)

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	epx	Epp/a	ib	Io	i(surge)	TE
Unit:	V	v	---	ma	mAdc	ma	°C
Maximum (5839):	27.8	1,375	---	230	75.0	750	160
Minimum (5839):	25.2	---	---	---	---	---	---
Maximum (5852):	6.6	1,375	---	230	75.0	750	160
Minimum (5852):	6.0	---	---	---	---	---	---
TEST CONDITIONS:							
(5839):	26.5	---	400	---	---	---	---
(5852):	6.3	---	400	---	---	---	---

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Zpp/a	tk	Rf	C:	Ehk	Alt
Unit:	Ohms	sec	Ohms	µF	v	ft
Maximum (5839):	Note 1	60	---	---	450	60,000 (See note 2)
Minimum (5839):	---	---	---	---	---	---
Maximum (5852):	Note 1	60	---	---	450	60,000 (See note 2)
Minimum (5852):	---	---	---	---	---	---
TEST CONDITIONS:						
(5839):	---	---	5,700	8	450	---
(5852):	---	---	5,700	8	450	---

GENERAL:

(D) Qualification - Not required
Reliable tubes
Stabilization (t = 45 hours) - See note 3

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS NOM	LIMITS MAX	UNIT
	Quality conformance inspection, part 1									
1301	Heater current									
	Type 5839	-		0.65	II	If	0.240	0.255	0.275	A
	Type 5852	-		0.65	II	If	1.100	1.200	1.300	A
1336	Heater-cathode leakage (1)	-	Ehk = 450 Vdc	0.65	II	Ihk	---	---	25	~~μAdc~~
1353	Operation of rectifiers	4,5		0.65	II	Io	70.0	75.0	---	mAdc
1231	Emission	-	E1b = 50 V; E2b = 0; E2b = 50 V; E1b = 0	0.65	II	I1s	120	---	---	mAdc
						I2s	120	---	---	mAdc
1201	Short and discontinuity detection	-		0.4	II	---	---	---	---	---
	Quality conformance inspection, part 2									
1031	Low-frequency vibration	-	No voltages	6.5	Code G	---	---	---	---	---
(D) 1031	Variable-frequency vibration	6	No voltage..	6.5	---	---	---	---	---	---
(D) 1031	Vibration fatigue	6		6.5	---	---	---	---	---	---
(D) ---	Post-vibration fatigue-test end points:	-								
1353	Operation of rectifiers	-	Ehk = 450 Vdc	---	---	Io	65.0	---	---	mAdc
1336	Heater-cathode leakage (1)	-	Ehk = 450 Vdc	---	---	Ihk	0	---	50	μAdc
1336	Heater-cathode leakage (2)	7,8	Ehk = 1,500 Vdc	6.5	II	Ihk	---	---	100	μAdc
1211	Insulation of electrodes									
	Type 5839	9	Ef = 26.5 Vdc; Ep = 1,500 Vdc	2.5	S3	---	---	---	---	---
	Type 5852	9	Ef = 6.3 Vdc; Ep = 1,500 Vdc							
(D) 1002	Barometric pressure, reduced	2,6	Pressure = 54.1 mmHg; 60,000 Ft	1.5	---	---	---	---	---	---
1101	Secureness of base, cap, or insert	-		---	---	---	---	---	---	---
1111	Base pin solder depth	-		---	---	---	---	---	---	---
1105	Permanence of marking	-		---	---	---	---	---	---	---

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NOTES: -Continued

3. Operation shall be approximately equivalent to test conditions and under substantially sinusoidal vibration at 30 ±5 cycles and 0.012-inch (0.30 mm) minimum displacement.

4. This test to be performed at the conclusion of the holding period.

5. In a full wave circuit, adjust Zp/a so that a tube having Etd = 23 Vdc at 70 mAdc per anode gives Io = 75 mAdc.

(D) 6. This test shall be conducted on the initial lot and thereafter on a lot approximately every 6 months. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes. When one lot has passed, the 6-month rule shall apply. MIL-STD-105, sample size code letter E, shall apply.

7. Time of test 1 hour. No arc-over or failure of heater or insulator permitted. At the end of the test, Ihk shall not exceed the specified value.

8. Heater positive with respect to cathode with 2 megohms in series with the source.

9. Heater and cathode tied together and positive with respect to anode.

10. A cycle shall be 2-1/2 minutes "on" and 2-1/2 minutes "off".

11. Values of R_L and C_L given in test conditions may be considered as approximate and shall be adjusted initially to give Io equal to, or greater than, Io = 75 mAdc with Ib equal to, or greater than, 230 ma. Ehk = Eo.

12. Tubes shall be life tested and accepted or rejected in accordance with the following chart.

Lot size	Life sample size	Total failures including inoperatives		Inoperatives only	
		Accept	Reject	Accept	Reject
0 - 1,000	15	2	3	1	2
1,001 - 3,000	25	3	4	1	2
3,001 - 9,000	40	4	5	2	3
9,001 - up	50	5	6	2	3

A failure is counted if a tube fails one or more of the life-test end points. Any tubes which show as air tubes, have a broken or cracked bulb, open heater or other open circuit, short, or have broken pins, are considered inoperative.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 17, 80
DSA - ES

User activities:
Army - MC
Navy - AS, MC, CG, SH

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-2912)

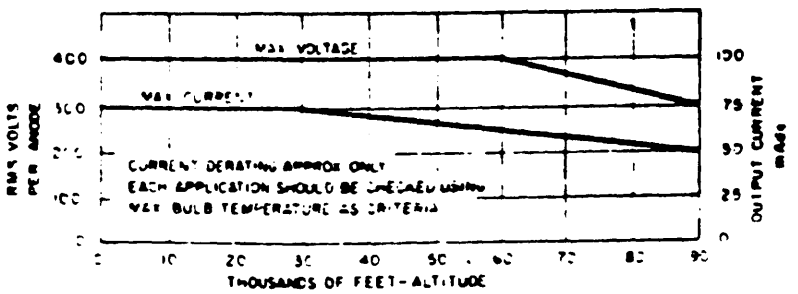
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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3								
1506	Heater-cycling life								
	Type 5839	10	Ef = 29.0 Vac; Ehk = 400 Vac	---	---	---	3,000	---	Cycles
	Type 5852	10	Ef = 7.0 Vac; Ehk = 400 Vac	---	---	---	3,000	---	Cycles
---	Heater-cycling life-test end points:								
	Type 5839								
1301	Heater current	-		---	---	If	0.230	0.315	A
1336	Heater-cathode leakage (2)	-		---	---	Ihk	---	100	uAdc
	Type 5852								
1301	Heater current	-		---	---	If	1.05	1.35	A
1336	Heater-cathode leakage (2)	-		---	---	Ihk	---	100	uAdc
1501	Intermittent life	11,12		---	---	---	---	---	---
---	Intermittent life-test end points (1,000 hours):								
	Type 5839								
1301	Heater current	-		---	---	If	0.230	0.315	A
1353	Operation of rectifiers	-		---	---	Io	65.0	---	mAdc
1336	Heater-cathode leakage (1)	-		---	---	Ihk	---	50	uAdc
1231	Emission	-		---	---	Ils	75.0	---	mAdc
1231	Emission	-		---	---	Il2s	75.0	---	mAdc
	Type 5852								
1301	Heater current	-		---	---	If	1.050	1.350	A
1353	Operation of rectifiers	-		---	---	Io	65.0	---	mAdc
1336	Heater-cathode leakage (1)	-		---	---	Ihk	---	50	uAdc
1231	Emission	-		---	---	Ils	75.0	---	mAdc
1231	Emission	-		---	---	Il2s	75.0	---	mAdc

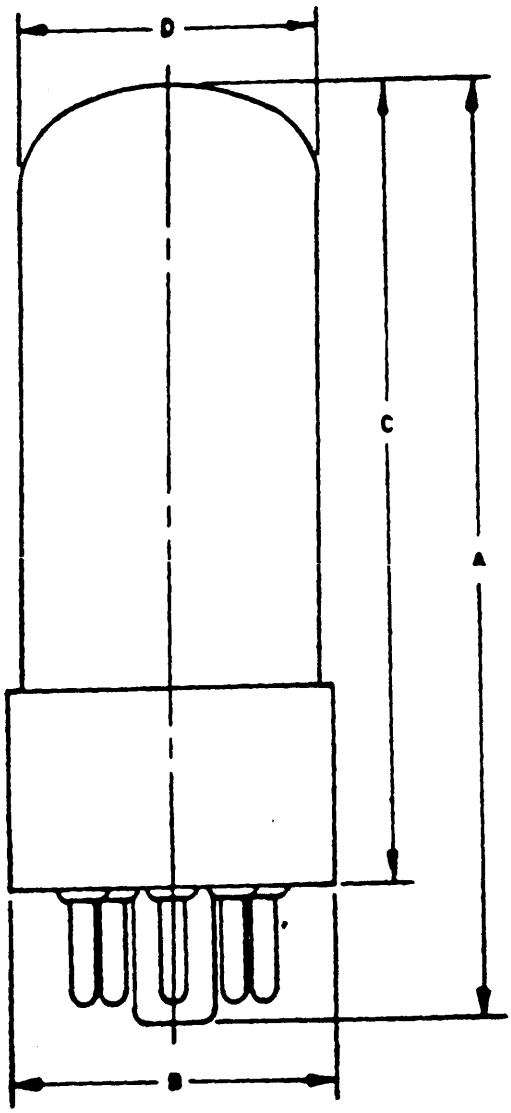
NOTES:

1. In order to insure that neither the peak anode current rating nor the surge current ratings are exceeded, the anode supply impedance per anode should be 150 ohms minimum. To insure that this condition is met in all designs, a resistor of between 100 and 150 ohms should be inserted in each anode lead.

2. The chart below indicates maximum rms anode voltage per anode and maximum dc anode current versus altitude ratings up to 90,000 feet.



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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 1		
A		3.320(84.33)
B		1.328(33.73)
C		2.820(71.63)
D	1.125(28.58) nominal	

FIGURE 1. Outline drawing of electron tube types 5830 and 5852.

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